

ANSWERS

$$\begin{aligned}\textcircled{1} \text{ S.A cube} &= (L \times W) \times 6 \\ &= (4 \times 4) \times 6 \\ &= 16 \times 6 \\ &= 96 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\textcircled{2} \text{ Vol of cube} &= L \times W \times h \\ &= 4 \times 4 \times 4 \\ &= 64 \text{ cm}^3\end{aligned}$$

$$\begin{aligned}\textcircled{3} \text{ S.A rectangle prism} &= (L_1 \times W_1) \times 2 + (L_2 \times W_2) \times 2 + (L_3 \times W_3) \times 2 \\ &= (8 \times 2) \times 2 + (8 \times 4) \times 2 + (4 \times 2) \times 2 \\ &= 16 \times 2 + 32 \times 2 + 8 \times 2 \\ &= 32 + 64 + 16 \\ &= 112 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\textcircled{4} \text{ Volume} &= L \times W \times h \\ &= 8 \times 2 \times 4 \\ &= 64 \text{ cm}^3\end{aligned}$$

$\textcircled{5}$ The volume of both solids is the same, but rectangular prism has 1.75 times greater surface area.

$\textcircled{7}$ Rectangular prism are the most wasteful. Cubes use less paper.

$\textcircled{8}$ - Harder to access on shelves due to putting in rows on top of each other.
- Two hands to pour due to width.
- Less S.A. for advertising & companies want a bigger box to stand out.